

ARK-445 Preregistration: Tri-State Authorization Discrimination (ALLOW / HOLD / DENY)

Experiment ID: ARK-445

Series: Authorization Boundary Characterization (Phase 1)

Preregistration Date: 2026-07-17 (UTC)

Execution Status: PENDING (preregistered, not yet executed)

Protocol Version: 28-field structured preregistration

Field 1: Central Research Question

Can the system reliably discriminate between **THREE** distinct authorization states — **ALLOW** (execute), **HOLD** (ambiguous/pending), and **DENY** (refuse) — rather than only binary **ALLOW/DENY**?

Specifically: Does the verify-then-execute (VBE) authorization boundary support a tri-state decision space where: - **ALLOW** -> payload executes with high confidence ($P(\text{execute}) \geq 0.90$), - **DENY** -> payload refuses with high confidence ($P(\text{execute}) \leq 0.02$ corrected), - **HOLD** -> payload exhibits intermediate behavior, distinguishable from both **ALLOW** and **DENY** ($P(\text{execute})$ in $[0.40, 0.60]$),

... and the three states are metrologically separable with margin $\Delta_{\text{H}} \geq 0.30$?

Field 2: Primary Hypothesis (H1)

H1: The system can discriminate **ALLOW**, **HOLD**, and **DENY** authorization states with clear separation: - **ALLOW arms** (authorizer $A=|1\rangle$): $S_{\text{A}} \geq 0.90$ (SPAM-corrected) - **DENY arms** (authorizer $A=|0\rangle$): $L_{\text{D_corrected}} \leq 0.02$ - **HOLD arms** (authorizer $A=|+\rangle$ or $|-\rangle$): H in $[0.40, 0.60]$ - **Tri-state discrimination margin:** $\Delta_{\text{H}} = \min(S_{\text{A_min}} - H_{\text{max}}, H_{\text{min}} - L_{\text{D_max}}) \geq 0.30$

where: - $S_{\text{A_min}}$ = minimum observed $P(Q_{\text{P}}=1)$ across all **ALLOW** arms (SPAM-corrected) - $H_{\text{max}}, H_{\text{min}}$ = max/min observed $P(Q_{\text{P}}=1)$ across all **HOLD** arms - $L_{\text{D_max}}$ = maximum observed $L_{\text{D_corrected}}$ across all **DENY** arms

Field 3: Secondary / Exploratory Hypotheses

H2a (Superposition symmetry): **HOLD** arms with $A=|+\rangle$ and $A=|-\rangle$ yield statistically indistinguishable $P(\text{execute})$, confirming basis-independent

ambiguity encoding.

H2b (Reverification escape): An ALLOW arm with delay + re-verification maintains $S_A \geq 0.90$, demonstrating the HOLD state is not an artifact of decoherence but a genuine intermediate encoding.

H2c (Confusion falls to DENY): Tampered or replayed authorization attempts collapse to the DENY outcome ($L \leq 0.02$ corrected), not to HOLD.

H2d (SPAM-drift bound): In-situ SPAM baselines for Q_A and Q_P remain ≤ 0.02 per qubit, and $|SPAM_A - SPAM_P| \leq 0.005$ (drift check).

Field 4: Null Hypothesis (H0)

H0: The system cannot reliably separate HOLD from ALLOW or DENY — either: 1. HOLD arms collapse into the ALLOW region ($H > 0.80$), or 2. HOLD arms collapse into the DENY region ($H < 0.10$), or 3. HOLD arms are statistically indistinguishable from SPAM baseline ($|H - SPAM| < 0.05$), or 4. Discrimination margin $\Delta_H < 0.30$ (insufficient separation for practical tri-state use).

Implication if H0 holds: The VBE boundary is fundamentally binary (ALLOW/DENY only), and intermediate/pending states are not metrologically accessible.

Field 5: Experiment Design & Methodology

5.1 Circuit Architecture

- **2-qubit system:** Q_A (authorizer), Q_P (payload execution indicator)
- **10 arms** (single QuantumCircuit with parameter sweep):

Arm	Label	Q_A Preparation	Q_P Conditioning	Expected Outcome
1	allow_standard	X on A	$ 1\rangle$	if $A=1$: X on P
2	deny_standard	I on A	$ 0\rangle$	if $A=1$: X on P
3	hold_plus	H (-> X on A)	$ +\rangle$	if $A=1$: X on P
4	hold_minus	X + H (-> X on A)	$ -\rangle$	if $A=1$: X on P
5	allow_alt	X on A	if $A=1$: X on P	ALLOW: S_A_alt
6	deny_alt	I on A	if $A=1$: X on P	DENY: L_D_alt
7	allow_reverification	X on A , delay 1 us, re-verify	if $A=1$: X on P	ALLOW: S_A_rev
8	deny_explore	X on A , delay 1 us	if $A=1$: X on P	DENY: L_D_exp
9	confusion_replay	X on A , delay 1 us, replay > discard, reset, I on A	if $A=1$: X on P	DENY: L_conf

Arm	Label	Q_A Preparation	Q_P Conditioning	Expected Outcome
10	spam_id1d		I (no gate)	SPAM: baseline

Key mechanism: - ALLOW: A prepared in $|1\rangle \rightarrow$ measurement collapses to outcome=1 $\rightarrow$ X fires on P $\rightarrow$ P read as $|1\rangle$ - DENY: A prepared in $|0\rangle \rightarrow$ measurement collapses to outcome=0 $\rightarrow$ X does NOT fire $\rightarrow$ P read as $|0\rangle$ - HOLD: A prepared in $|+\rangle$ or $|-\rangle \rightarrow$ measurement collapses to 0 or 1 with $\sim 50\%$ probability EACH shot $\rightarrow$ aggregate $P(Q_P=1) \sim 0.5$ across shots (intermediate between 0 and 1)

Classical feedforward: Measure Q_A $\rightarrow$ ClassicalRegister `ca`, condition X on Q_P using single-register `if_test(ca==1, ...)` (flat condition, no nested blocks — ARK-444 error-1524 lesson applied).

5.2 Qubit Selection (Execution-Time)

- **Rule:** 2 qubits (Q_A, Q_P) with:
 1. Readout error (RE) $< 2\%$ for BOTH qubits
 2. Q_A and Q_P are **connected** (CNOTs allowed, though not used in this experiment)
 3. **Minimum sum:** $\text{argmin}(\text{RE}_A + \text{RE}_P)$ across all valid connected pairs
 4. Selected from **live calibration data** on execution day (not preregistered)
 5. `initial_layout = [Q_A, Q_P]` enforced in transpile
- **Selection frozen** at execution and committed to `selected_qubits.json` BEFORE any hardware job.

5.3 Execution Parameters

- **Backend:** `ibm_marrakesh` (156-qubit Heron r2, preferred); fallback `ibm_fez` if marrakesh unavailable
- **Shots per arm:** 8,192
- **Total shots:** 81,920 (10 arms x 8,192)
- **Transpiler:** `optimization_level=3`, `seed_transpiler=445`, `initial_layout=[Q_A, Q_P]`, no dynamical decoupling (`None`)
- **Sampler:** `SamplerV2` (shots mode), `default_shots=8192`
- **Estimated runtime:** $\sim 25\text{--}30$ seconds (principal job)

Field 6: Measured Observables & Metrics

Primary Metrics

All metrics computed from raw counts (no readout mitigation):

1. **S_A (ALLOW arms 1, 5, 7):** $P(Q_P=1 \mid \text{ALLOW})$ for each arm
 - $S_A_min = \min(S_A_arm1, S_A_arm5, S_A_arm7)$
2. **L_D (DENY arms 2, 6, 8, 9):** $P(Q_P=1 \mid \text{DENY})$ for each arm
 - $L_D_corrected = \max(0, L_D - \text{SPAM_P})$ for each DENY arm
 - $L_D_max = \max(L_D_corrected_arm2, \dots, L_D_corrected_arm9)$
3. **H (HOLD arms 3, 4):** $P(Q_P=1 \mid \text{HOLD})$ for each arm
 - $H_plus = P(Q_P=1 \mid \text{arm3_hold_plus})$
 - $H_minus = P(Q_P=1 \mid \text{arm4_hold_minus})$
 - $H_min = \min(H_plus, H_minus)$
 - $H_max = \max(H_plus, H_minus)$
4. **SPAM baselines (arm 10):**
 - $\text{SPAM_A} = P(Q_A=1 \mid \text{idle})$
 - $\text{SPAM_P} = P(Q_P=1 \mid \text{idle})$
 - $\text{SPAM_drift} = |\text{SPAM_A} - \text{SPAM_P}|$
5. **Tri-state discrimination margin:**
 - $\Delta_H = \min(S_A_min - H_max, H_min - L_D_max)$

Diagnostic Metrics

- **I_H (HOLD symmetry index):** $|H_plus - H_minus|$ (expect < 0.05 if symmetric)
 - **L_conf (confusion/replay arm 9):** $P(Q_P=1 \mid \text{confusion})$ — expect ≤ 0.02 corrected
 - **S_A_rev (reverified ALLOW arm 7):** expect ≥ 0.90 (HOLD is not decoherence)
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Field 7: Success Criteria (Quantitative Thresholds)

PASS Conditions (ALL must be satisfied):

1. **ALLOW floor:** $S_A_min \geq 0.90$ (at least one ALLOW arm corrected if needed)
2. **DENY ceiling:** $L_D_corrected \leq 0.02$ for ALL four DENY arms
3. **HOLD range:** $0.40 \leq H_min$ AND $H_max \leq 0.60$ for BOTH HOLD arms
4. **Tri-state margin:** $\Delta_H \geq 0.30$
5. **SPAM kill-gate (pre-check):** arm10 $\text{SPAM_A} \leq 0.02$ AND $\text{SPAM_P} \leq 0.02$ (executed BEFORE principal job; principal job cancelled if violated)

FAIL Conditions (any one triggers FAIL):

1. $S_A_min < 0.90$ after SPAM correction
2. ANY $L_D_corrected > 0.02$
3. $H_min < 0.40$ OR $H_max > 0.60$ (HOLD collapsed to ALLOW or DENY)

4. $\Delta_{H} < 0.30$ (insufficient discrimination)

KILL / INDETERMINATE Conditions:

1. **SPAM gate violation:** $\text{SPAM_A} > 0.02$ OR $\text{SPAM_P} > 0.02$ from arm10 -> **KILL** experiment immediately, do not submit principal job
 2. **Drift violation:** $\text{SPAM_drift} > 0.005$ -> **INDETERMINATE**
 3. **Job failure:** Principal job returns error, zero counts, or incomplete data -> **INDETERMINATE**
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Field 8: Statistical Power & Sample Size

- **Shots per arm:** 8,192
 - **Binomial SEM** at $p=0.50$: $\sigma \sim \sqrt{0.5 \times 0.5 / 8192} \sim 0.0055$ (0.55%)
 - **95% CI half-width:** $\pm 1.96\sigma \sim \pm 0.011$ ($\pm 1.1\%$)
 - **Detectability:** HOLD range $[0.40, 0.60]$ is $\pm 10\%$ from ideal 0.50, well above $\pm 1.1\%$ noise -> robustly detectable
 - **Discrimination margin 0.30** (30 percentage points) » 2% noise -> clear separation
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Field 9: Planned Statistical Analysis

Analysis Steps:

1. **Raw count extraction:** Parse SamplerV2 results for each arm
2. **SPAM correction:** For DENY arms, compute $L_D_corrected = \max(0, L_D_raw - \text{SPAM_P})$
3. **Metric computation:** Calculate S_A , L_D , H , Δ_{H} per Field 6 definitions
4. **Criterion scoring:** Evaluate PASS/FAIL per Field 7 thresholds
5. **Secondary hypothesis tests:**
 - H2a: two-sample proportion test for H_plus vs H_minus (expect $p > 0.05$)
 - H2b: verify $S_A_rev \geq 0.90$
 - H2c: verify $L_conf \leq 0.02$ corrected
 - H2d: verify $\text{SPAM_drift} \leq 0.005$

Outputs:

- **proofrecord.json:** all raw/corrected metrics + verdict + timestamp
- **plots/arm_results.png:** bar chart of $P(Q_P=1)$ for all 10 arms with thresholds
- **plots/tristate_discrimination.png:** scatter plot showing ALLOW/HOLD/DENY regions

Field 10: Interpretation Boundaries

What this experiment DOES test:

- Whether three authorization states (ALLOW/HOLD/DENY) can be metrologically distinguished on this specific 2-qubit setup with this circuit design.
- Whether superposition-based encoding ($|+\rangle$, $|-\rangle$) produces a stable intermediate outcome distinct from basis states.

What this experiment DOES NOT claim:

- **Not new physics:** Superposition $\rightarrow$ probabilistic measurement outcome is textbook quantum mechanics. This is a metrological characterization, not a discovery.
- **Not a security guarantee:** HOLD state accessibility does not imply cryptographic security, tamper evidence, or resistance to any attack model.
- **Not generalizable:** Results apply to the selected qubits on the selected backend on the execution date. No claims about other qubits, backends, or gate implementations.
- **Not a primitive:** This does not define or validate a “quantum tri-state authorization protocol” for real-world use.

Field 11: Dependencies on Prior Experiments

- **ARK-441/ARK-446:** Established binary ALLOW/DENY boundary ($\Delta_{B \geq 0.70}$, $L_D \leq 0.02$, $S_A \geq 0.90$). ARK-445 extends this to a tri-state space.
- **ARK-442:** Characterized delay-induced degradation. ARK-445 arm7 (reverified ALLOW) uses 1us delay to confirm HOLD is not a decoherence artifact.
- **ARK-444:** Demonstrated integrity gate (tamper $\rightarrow$ DENY). ARK-445 arm9 (confusion/replay) confirms tamper collapses to DENY, not HOLD.
- **ARK-443:** Validated classical feedforward (single-register `if_test`). ARK-445 uses the same flat-condition pattern (no nested blocks).

Execution order: ARK-441 $\rightarrow$ ARK-446 $\rightarrow$ ARK-442 $\rightarrow$ ARK-444 $\rightarrow$ ARK-443 $\rightarrow$ **ARK-445** (current) $\rightarrow$ ARK-447 (planned).

Field 12: Equipment & Resources

- **Platform:** IBM Quantum (`ibm_quantum_platform`), open instance
- **Backend:** `ibm_marrakesh` (156-qubit Heron r2) — preferred

- Fallback: `ibm_fez` (156-qubit Heron r2) if marrakesh offline
 - **Calibration data:** Live backend properties fetched on execution day
 - **Qiskit version:** `qiskit-ibm-runtime` ≥ 0.30 , `qiskit` ≥ 1.3
 - **Estimated usage:** $\sim 25\text{--}30\text{s}$ (principal) + $\sim 6\text{s}$ (SPAM) = $\sim 31\text{--}36\text{s}$ total
 - **Budget check:** Confirmed $\geq 40\text{s}$ remaining before submission
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Field 13: Data Collection Procedures

1. **Qubit selection:** Execute `ark_445_select_qubits.py` -> freeze selection in `selected_qubits.json` + calibration snapshot -> commit
 2. **SPAM kill-gate:** Execute `ark_445_spam_job.py` (arm10 only) -> wait for completion -> record SPAM results in `spam_results.json` -> **commit SPAM gate BEFORE principal** -> KILL if violated, else proceed
 3. **Principal job submission:** Execute `ark_445_submit_ibm.py` (arms 1–9) -> record job ID in `principal_job_id.txt` + metadata in `principal_job_meta.json` -> **commit job ID record BEFORE** retrieving results
 4. **Result retrieval:** Execute `ark_445_retrieve.py` -> poll job -> write `raw_results.json` -> commit
 5. **Analysis:** Execute `ark_445_analysis.py` -> compute metrics -> write `proofrecord.json` + plots -> commit
 6. **RESULTS:** Populate `RESULTS.md` -> generate `.docx` / `.pdf` -> commit + tag `ark-445-v1.0`
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Field 14: Planned Data Transformations

- **No readout error mitigation** (results reported from raw counts)
 - **SPAM correction:** For DENY arms only: $L_D_corrected = \max(0, L_D_raw - SPAM_P)$
 - **Aggregation:** All metrics aggregate counts within each arm (no cross-arm averaging)
 - **Filtering:** None (all shots included)
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Field 15: Exclusion Criteria

Pre-execution exclusions (cancel experiment): 1. No connected qubit pair with $RE < 2\%$ for both qubits available 2. SPAM kill-gate violated ($SPAM_A > 0.02$ OR $SPAM_P > 0.02$)

Post-execution exclusions (classify as INDETERMINATE): 1. Principal job fails (error code, zero counts, timeout) 2. $SPAM_drift > 0.005$ 3. Backend suffers mid-job recalibration or downtime

No per-shot exclusions (outlier shots are not filtered).

Field 16: Missing Data Handling

- **Incomplete job counts:** If any arm has $< 8,192$ shots returned -> INDETERMINATE
 - **Missing SPAM baseline:** If arm10 fails or returns zero counts -> KILL (cannot correct)
 - **Partial job failure:** If principal job completes but ≥ 1 arm missing -> INDETERMINATE
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Field 17: Exploratory Analyses (Post-Hoc, Not Pre-Committed)

After primary analysis, MAY explore (not binding on PASS/FAIL): - Shot-by-shot correlation between Q_A measurement outcome and Q_P readout in HOLD arms - Comparison of H_plus and H_minus to ideal 0.50 (Poisson test) - Cross-arm variance in S_A and L_D (robustness check)

These are descriptive only — no exploratory result can change the PASS/FAIL verdict.

Field 18: Limitations & Potential Confounds

1. **Decoherence vs encoding:** HOLD arms use superposition -> measurement-induced collapse. If T1/T2 decay is significant (unlikely at $< 1\mu s$), H might drift from 0.50. Arm7 (reverified ALLOW with $1\mu s$ delay) controls for this.
 2. **Gate fidelity:** If Hadamard gate fidelity is poor, $|+\rangle/|-\rangle$ preparation may be imperfect -> H drifts toward basis-state outcomes. No mitigation; reported as-observed.
 3. **Transpiler reordering:** `if_test` blocks are fragile to transpilation. Pre-lock validation (test transpile on real backend) will confirm correct gate sequence.
 4. **Endianness:** ClassicalRegister bit-order matters for `if_test(ca==1)`. Tested in pre-lock validation.
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Field 19: Deviations from Pre-Registration

Any deviation from this protocol (parameter changes, arm redefinition, threshold adjustment, additional SPAM correction) MUST be: 1. Documented in

RUN_LOG.md with justification 2. Committed BEFORE reading principal job results 3. Flagged in RESULTS.md as a protocol deviation

No post-data rescue is permitted. If primary hypotheses fail, the experiment fails — no threshold adjustment, no arm redefinition, no re-analysis can change the verdict.

Field 20: Preregistration Timestamp & Commitment

- **Preregistration completed:** 2026-07-17 (UTC)
 - **Committed to git:** execute/ark-445 branch
 - **LOCK commit (before any hardware job):** This preregistration file + all code files + MANIFEST.txt with SHA-256 hashes
 - **Public record:** Pushed to derekhone/executionproof-testbeds (public repo)
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Field 21: Roles & Responsibilities

- **Principal Investigator:** Derek Hone (Remnant Fieldworks Inc.)
 - **Experiment execution:** Automated via preregistered Python scripts
 - **Hardware provider:** IBM Quantum (ibm_marrakesh / ibm_fez)
 - **Code repository:** <https://github.com/derekhone/executionproof-testbeds>
 - **Funding:** None (open-access IBM Quantum)
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Field 22: Conflicts of Interest

None declared. This is an independent metrological characterization with no commercial interest, patent application, or external funding.

Field 23: Ethical Considerations

- **No human subjects, biological materials, or animal use**
 - **No dual-use concern:** Quantum gate characterization; no cryptographic claims
 - **Environmental:** Negligible energy (~30s compute on shared IBM hardware)
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Field 24: Data Sharing & Availability

All data will be made publicly available upon completion: - Pre-registration (this file), code, MANIFEST committed BEFORE execution
- `selected_qubits.json` with live calibration data - `spam_results.json` (SPAM kill-gate) - `principal_job_id.txt`, `principal_job_meta.json` (job provenance) - `raw_results.json` (raw counts from IBM backend) - `proofrecord.json` (all metrics + verdict) - RESULTS.md / .docx / .pdf (human-readable report) - Plots: `arm_results.png`, `tristate_discrimination.png`

Repository: <https://github.com/derekhone/executionproof-testbeds/tree/execute/ark-445>

Release tag: ark-445-v1.0 (created upon completion)

Zenodo DOI: Will be updated in record 21398676 (ARK-441 dataset continuation)

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Field 25: Timeline

- **Preregistration lock:** 2026-07-17 (this document + code + MANIFEST committed)
- **Qubit selection:** 2026-07-17 (execution day, after lock)
- **SPAM gate:** 2026-07-17 (before principal job)
- **Principal job submission:** 2026-07-17 (after SPAM gate passes)
- **Analysis & results:** 2026-07-17 (same day)
- **Public release:** 2026-07-17 (tag + PR opened)

Total elapsed time (preregistration -> results): < 2 hours (same-day execution)

Field 26: Contingency Plans

If SPAM kill-gate fails (SPAM > 0.02):

- **Action:** KILL experiment, do NOT submit principal job
- **Outcome:** Report as INDETERMINATE (hardware SPAM too high for reliable measurement)
- **Retry:** MAY retry on a different day or backend (new preregistration freeze required)

If principal job fails (error, zero counts, timeout):

- **Action:** Classify as INDETERMINATE, do NOT resubmit
- **Outcome:** Report job failure in RESULTS.md

- **No rescue:** Failure is a valid outcome; no retry within this preregistration instance

If backend goes offline mid-execution:

- **Action:** Report as INDETERMINATE
- **No fallback mid-execution:** Cannot switch backends after LOCK (qubits are backend-specific)

Field 27: Ordering Constraint (Strict Sequence)

Hard ordering to prevent post-hoc bias or rescue-after-failure:

1. LOCK (this prereg + code + MANIFEST) -> commit SHA recorded in RUN_LOG.md
2. Qubit selection (selected_qubits.json + calibration snapshot) -> commit
3. SPAM job submission (arm10 only) -> wait for completion
4. SPAM gate result (spam_results.json) -> commit BEFORE principal job
5. [KILL-GATE] IF SPAM violated -> STOP, classify INDETERMINATE, commit RUN_LOG, DONE
6. [IF SPAM passes] -> Principal job submission (arms 1-9)
7. Principal job ID record (principal_job_id.txt + metadata) -> commit BEFORE reading results
8. Result retrieval (wait for job completion)
9. Raw results (raw_results.json) -> commit
10. Analysis (proofrecord.json + plots) -> commit
11. RESULTS.md + docx/pdf -> commit
12. Tag ark-445-v1.0 -> push all commits + tag
13. Open PR (do NOT merge) -> request review

Enforcement: Each step writes files + git commit. Commit timestamps prove ordering. Any violation (e.g., job ID committed after results) invalidates the experiment.

Field 28: Amendment Log

- **Version 1.0 (2026-07-17):** Initial preregistration — LOCKED, no amendments

Any future amendments MUST: 1. Be committed BEFORE the step they affect 2. Be justified in RUN_LOG.md with rationale 3. Preserve original text (amendments append, do not overwrite)

Post-LOCK, pre-data amendments are permissible ONLY for technical corrections (e.g., transpiler syntax error discovered in pre-execution test). Post-data amendments are FORBIDDEN.

Summary: Quick Reference

Parameter	Value
Qubits	2 (Q_A, Q_P) — execution-time selection, RE < 2%, connected
Arms	10 (3 ALLOW, 4 DENY, 2 HOLD, 1 SPAM)
Shots/arm	8,192
Backend	ibm_marrakesh (preferred) / ibm_fez (fallback)
PASS rule	S_A >= 0.90, L_D <= 0.02, H in [0.40,0.60], Delta_H >= 0.30, SPAM <= 0.02
KILL rule	SPAM > 0.02 (any qubit) -> cancel principal job
Central claim	Tri-state authorization (ALLOW/HOLD/DENY) is metrologically discriminable
Interpretation	Metrological characterization only — not new physics, not cryptographic security

END OF PREREGISTRATION (28 fields complete)

This document serves as the LOCK for ARK-445. Commit SHA + timestamp
in RUN_LOG.md will timestamp the preregistration freeze.